

Danmarks
Tekniske
Universitet



Technical University of Denmark (DTU)

Department of Applied Mathematics and Computer Science (DTU Compute)

MASTER'S THESIS PROJECT TIME PLAN

Explainable Self-Supervised Network Anomaly Detection for Cyber Defense Using Multi-Modal Network Telemetry

Student Name: Mounika Maidamshetti

Student ID: S250148

Programme: MSc Computer Science and Engineering

Department: DTU Compute

1 Research Goal

The goal of this thesis is to develop and evaluate an explainable self-supervised anomaly detection framework for cyber defence. The system will learn normal network behaviour from largely unlabelled network telemetry and use this knowledge to identify previously unseen or suspicious activity. Self-supervised learning (SSL) offers a promising approach by learning meaningful patterns from large amounts of unlabeled network telemetry, reducing the dependence on labeled attack data. Many AI-based anomaly detection systems lack transparency, restricting their usefulness in operational cybersecurity environments. This thesis proposes an explainable self-supervised network anomaly detection framework that uses multi-modal network telemetry to identify unknown cyber threats while providing interpretable explanations to support cybersecurity analysts.

The proposed framework is designed to operate on network telemetry collected from security monitoring platforms such as Suricata and Zeek, which generate flow, DNS, TLS, HTTP, protocol, and alert logs. These telemetry sources provide complementary information about network behaviour and form the basis for learning normal communication patterns in operational cyber defence environments.

2 Problem Statement

Current network intrusion detection systems primarily rely on predefined signatures or rule-based methods, supervised learning methods that require labeled attack data. These approaches are often ineffective against novel or developing threats and provide limited insight into why an event is classified as anomalous. There is therefore a need for an anomaly detection framework capable of learning normal network behaviour from unlabelled multi-modal telemetry while producing interpretable explanations for security analysts.

How can self-supervised learning and explainable AI be combined to detect previously unseen cyber threats from network telemetry while providing useful decision support for cybersecurity analysts?

3 Research Questions

- **RQ1:** How effectively can self-supervised learning detect previously unseen network attacks without labeled training data?
- **RQ2:** How does combining multiple telemetry sources improve anomaly detection performance?
- **RQ3:** Which Explainable AI methods provide meaningful explanations for anomaly detection decisions?

- **RQ4:** What trade-offs exist between detection performance and false positive reduction in SSL-based anomaly detection?
- **RQ5:** How suitable is the proposed approach for deployment within Terma’s cyber defense environments?

4 Learning Objectives

The primary objective is to design, implement, and evaluate an explainable self-supervised anomaly detection framework capable of identifying unknown cyber attacks from multi-modal network telemetry. Specific objectives include:

1. Understand existing network anomaly detection techniques and identify research gaps.
2. Gain knowledge of self-supervised learning methods for unlabelled network data.
3. Develop a multi-modal representation by combining different network telemetry sources.
4. Develop and train self-supervised models for network behaviour analysis.
5. Detect abnormal network behaviour using learned representations.
6. Integrate explainable AI techniques such as SHAP, LIME and feature attribution.
7. Evaluate the proposed framework using benchmark cybersecurity datasets.
8. Analyse deployment feasibility for real-world defence environments

5 Project Phases

1. Literature Review, Research Definition.
2. Select suitable datasets and understand Suricata/Zeek data sources and prepare the data collection and validation strategy.
3. Data preprocessing performs feature engineering and establish baseline anomaly detection models.
4. Design, implement, and train the proposed self-supervised anomaly detection framework.
5. Integrate XAI methods such as SHAP and LIME. Evaluate detection performance
6. Analyse results and limitations, and assess deployment feasibility in the defence system.
7. Revise the thesis, finalise the document and prototype.

6 Proposed Thesis Structure

- Introduction
- Background and Related Work
- Problem Definition and Research Questions
- Methodology and Dataset
- Proposed Framework
- Implementation
- Experimental Results
- Discussion and Limitations
- Conclusion
- Future Work

7 Expected Outcomes

The expected outcome of this thesis research is an explainable anomaly detection framework capable of identifying previously unknown cyberattacks using unlabeled network telemetry. The framework is expected to reduce false positives while providing meaningful explanations that improve analyst trust and support faster incident investigation. The results will contribute to the integration of trustworthy artificial intelligence into real-world cyber defense system and provide valuable insights for future deployment within industrial environments.

8 Conclusion

This thesis aims to develop and evaluate an explainable self-supervised anomaly detection framework for cyber defense. By combining SSL, multi-modal telemetry fusion, anomaly detection techniques, and explainable AI methods, the proposed system search aims to improve the detection of previously unseen attacks while reducing false positives and increasing analyst trust. The proposed framework is intended to complement existing signature-based detection systems, such as Suricata and Zeek, by providing an additional machine learning based anomaly detection capability rather than replacing existing security monitoring mechanisms. The expected outcome is a practical and explainable anomaly detection prototype that contributes to both academic research and industrial cybersecurity applications, particularly in mission-critical environments.

Table 1: Gantt chart for the thesis project timeline

Activity	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Literature review & research definition	■					
Dataset selection & telemetry understanding	■	■				
Data preprocessing & feature engineering		■	■			
Baseline anomaly detection models		■	■			
SSL framework design & implementation			■	■		
Model training & iterative optimization			■	■		
XAI integration (SHAP/LIME)				■	■	
Benchmark evaluation & comparison				■	■	
Results analysis & deployment feasibility					■	
Thesis writing & supervisor/company feedback					■	■
Final revision, presentation & defence						■